SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent
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COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Analytical Problem-Solving Assessment

A game-playing AI is implemented using DQN, Double DQN (DDQN), and Dueling DQN. Analyse the advantages of DDQN and Dueling DQN over the standard DQN architecture.

Ans: -

Introduction

A game-playing AI can use Deep Reinforcement Learning to learn the best actions by interacting with a game environment. Deep Q-Network (DQN) uses a neural network to estimate the value of possible actions. However, standard DQN has some limitations, especially Q-value overestimation and inefficient learning when many actions have similar effects.

Two improved approaches are Double DQN (DDQN) and Dueling DQN. Both improve the performance of standard DQN in different ways.

1. Deep Q-Network (DQN)

DQN uses a neural network to estimate the Q-value of each possible action.

Game State
↓
Neural Network
↓
Q-values
↓
Select Best Action

For example:
Left → 5
Right → 8
Jump → 10
The agent selects Jump because it has the highest Q-value.

Limitation of DQN
DQN can sometimes overestimate Q-values because the same process is used to select and evaluate the best action. This can result in unstable learning and a less accurate policy.

2. Double DQN (DDQN)
Double DQN is an improvement over standard DQN that reduces the overestimation problem.
It uses two networks:

- Online network – selects the best action.
- Target network – evaluates the selected action.

```
 Current State
      ↓
Online Network
      ↓
Select Best Action
      ↓
 Target Network
      ↓
 Evaluate Action
```

Advantages of DDQN

1. Reduces Q-value overestimation.
2. Provides more accurate action values.
3. Improves training stability.
4. Produces better policies in many environments.
5. Helps the agent make more reliable decisions.

3. Dueling DQN
Dueling DQN modifies the architecture of DQN by separating the estimation of state value and action advantage.
It has two streams:

```
        Game State
            ↓
      Shared Network
        ↙      ↘
 State Value    Advantage
    V(s)         A(s,a)
        ↘      ↙
        Q-values
```

State Value
It estimates how good the current state is, independent of a particular action.

Action Advantage
It estimates how much better one action is compared with other actions.

Advantages of Dueling DQN

1. Learns useful states more efficiently.
2. Identifies important actions.
3. Works well when several actions have similar outcomes.
4. Improves learning efficiency.
5. Can provide better performance in complex environments.

4. Comparison

Feature	**DQN**	**DDQN**	**Dueling DQN**
Basic architecture	Standard Q-network	Online + Target networks	Value + Advantage streams
Q-value overestimation	Higher	Reduced	Not its primary focus
Learning stability	Moderate	High	High
State-value estimation	Combined	Combined	Separate
Action advantage	Combined	Combined	Separate
Complex environments	Good	Better	Better
Main improvement	Basic deep Q-learning	Accurate Q-values	Better state/action representation

5. Example
Consider a game where the agent can perform:
Action 1 → Move Left
Action 2 → Move Right
Action 3 → Jump

Standard DQN may estimate:
Left = 5
Right = 8
Jump = 12
The agent selects Jump.
However, the value 12 may be an overestimated value.

DDQN
DDQN uses one network to select Jump and another network to evaluate it. This reduces the chance of selecting an action because of an incorrectly high Q-value.

Dueling DQN
Dueling DQN first determines whether the current game state is valuable and then determines which action provides the greatest advantage.

6. Advantages in Game Playing

Using DDQN and Dueling DQN can provide:

- Better decision-making
- More stable training
- Reduced estimation errors
- Improved learning efficiency
- Better performance in complex game environments
- More reliable action selection

7. Conclusion

Standard DQN provides a powerful method for learning game-playing policies but can suffer from Q-value overestimation. DDQN addresses this problem by separating action selection from action evaluation. Dueling DQN improves the network architecture by separately estimating the value of a state and the advantage of each action.
Therefore, DDQN is mainly useful for reducing overestimation, while Dueling DQN is mainly useful for improving state and action-value representation. Both can outperform standard DQN in suitable complex game environments.

Code of Conduct:

I U. Somesh Kumar (192425019) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U. Somesh Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively